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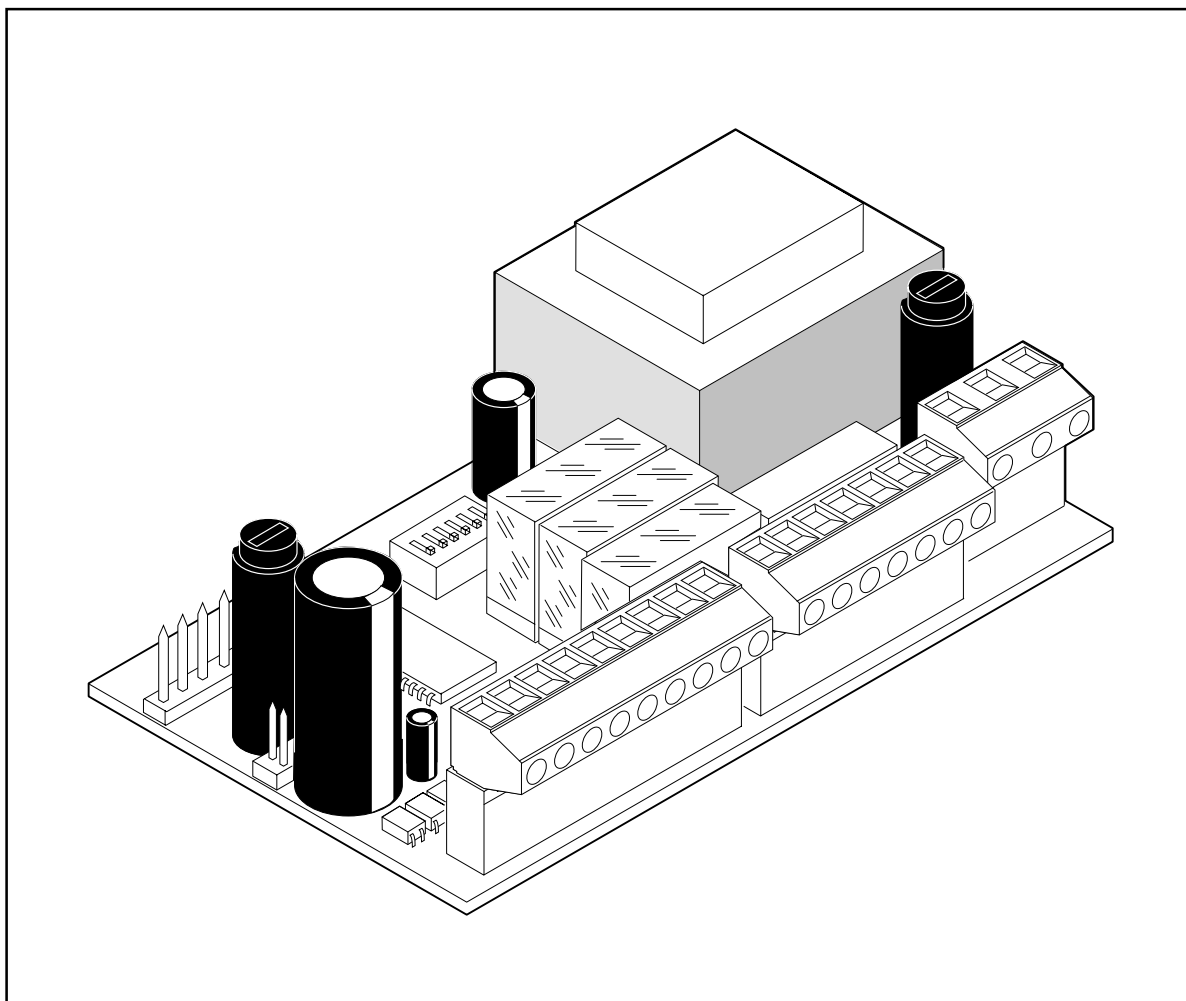
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596MPS & 610MPS



FAAC



UNI EN ISO 9001-085

CE DECLARATION OF CONFORMITY

Manufacturer: FAAC S.p.A.

Address: Via Benini, 1 - 40069 Zola Predosa BOLOGNA - ITALY

Declares that: Electronic appliances 596MPS and 610MPS

- conform to the essential safety requirements of the following directives:

73/23/EEC and subsequent amendment 93/68/EEC.

89/336/EEC and subsequent amendment 92/31/CEE and 93/68/EEC

Additional information:

These products underwent tests in a typical, uniform configuration (all products manufactured by FAAC S.p.A.).

Bologna 01 January 1999

The Managing Director
A. Bassi



WARNING FOR THE INSTALLER

GENERAL SAFETY OBLIGATIONS

- CAUTION! It is important for personal safety to follow all the instructions carefully. Incorrect installation or misuse of the product may cause people serious harm.**
- Read the instructions carefully before starting to install the product.
- Packaging material (plastic, polystyrene, etc.) must not be left within reach of children as it is a potential source of danger.
- Keep the instructions in a safe place for future reference.
- This product was designed and manufactured strictly for the use indicated in this documentation. Any other not expressly indicated use may damage the product and/or be a source of danger.
- FAAC accepts no responsibility due to improper use of the automation or use other than that intended.
- Do not install the appliance in an area subject to explosion hazard: inflammable gasses or fumes are a serious safety hazard.
- Mechanical construction elements must meet the provisions of UNI 8612, CEN prEN 12604 and CEN prEN 12605 Standards.
To obtain an adequate level of safety in non EU countries, the above mentioned Standards must be observed in addition to national standards.
- FAAC will not accept responsibility if the principles of Good Workmanship are disregarded in constructing the closing elements to be motorised, and if any deformation occurs during use of the said elements.
- Installation must meet the following Standards: UNI 8612, CEN prEN 12453 and CEN prEN 12635.
The safety class for the automation must be C+D.
- Before carrying out any work on the system, switch off the power supply.
- The mains power supply of the automation must be fitted with a all-pole switch with contact opening distance of 3mm or greater. Use of a 6A thermal breaker with all-pole circuit break is recommended.
- Make sure there is a differential switch with 0.03A threshold upstream of the system.
- Check that the earthing system is correctly made and connect the closure metal parts to it. Also connect the Yellow/Green wire of the automation to the earthing system.
- The automation includes an intrinsic anti-crushing device consisting of a torque control which, however, must be installed together with other safety devices.
- The safety devices [e.g.: photocells, sensitive edges, etc...] protect any dangerous areas against **Movement mechanical risks**, such as, crushing, dragging, or shearing.
- Use of at least one indicator-light is essential for each system (e.g.: FAAC LAMP MINILAMP, etc.) as well as a sign adequately fixed to the frame structure, in addition to the devices mentioned at point "16".
- FAAC accepts no responsibility regarding safety and correct operation of the automation, should components made by manufacturers other than FAAC be used in the system.
- Use only FAAC original spare parts for maintenance.
- Do not make any alterations to the components of the automation.
- The installer must supply full information regarding manual operation of the system in case of an emergency and hand to the user of the system the "User's Guide" included with the product.
- Do not allow children or other persons to stand near the product while in operation.
- Keep remote controls or any other pulse generator well away from children, to prevent the automation from being activated accidentally.
- The user must refrain from attempting to repair or adjust the system personally and should contact qualified personnel only.
- Anything not expressly provided for in these instructions is not permitted.**

596MPS & 610MPS

These instructions apply to the following models:

Appliances 596MPS - 610MPS

1. DESCRIPTION

The 596MPS electronic appliance is suitable for controlling up-and-over automatic systems with single or double hydraulic operator.

The 610MPS electronic appliance is suitable for controlling barriers with a hydraulic operator.

1.1. TECHNICAL SPECIFICATIONS

TABLE 1 OPERATING PARAMETERS OF 596MPS & 610MPS (dip-switch selectable)

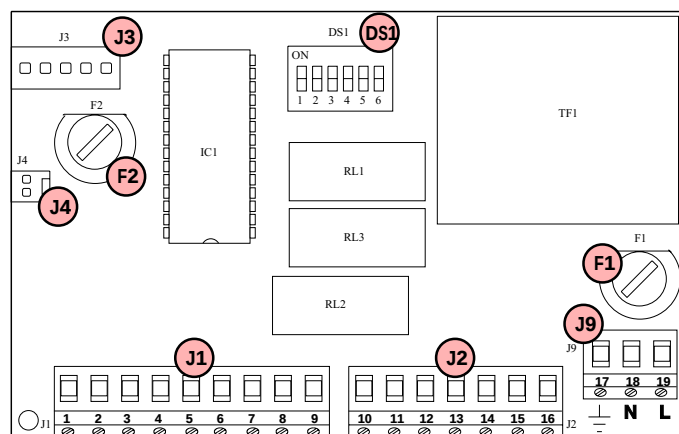
	596MPS	610MPS
Logic	automatic (A) / semi-automatic (E)	
Pause time (sec.)	30/60	0/2/5/10/15/30/60
Work time (sec.)	25/30/35/40	4/5/7/9
Failsafe	enabled/disabled	

TABLE 2 HARDWARE CHARACTERISTICS OF 596MPS & 610MPS

Power supply	230V~ (+6 -10 %) 50Hz
Max absorbed power	10 W
Motors max. load	800 W
Power supply for accessories	24Vdc
Accessories max. load	250 mA
Ambient temperature	- 20°C ÷ + 55°C
Fuses	motor/accessories
Rapid connectors	for decoding-RP receiver cards push-button for outside opening
Terminal boards	removable
Terminal board inputs	Open Stop Closure safety devices Limit-switch
Terminal board outputs	motor capacitor 24 Vdc accessories power supply courtesy light (596MPS) flashlight (610MPS)
Courtesy timing	90 sec. (596MPS)

1.2. LAY-OUT OF 596MPS & 610MPS CARD

Fig. 1



2. ELECTRICAL CONNECTIONS

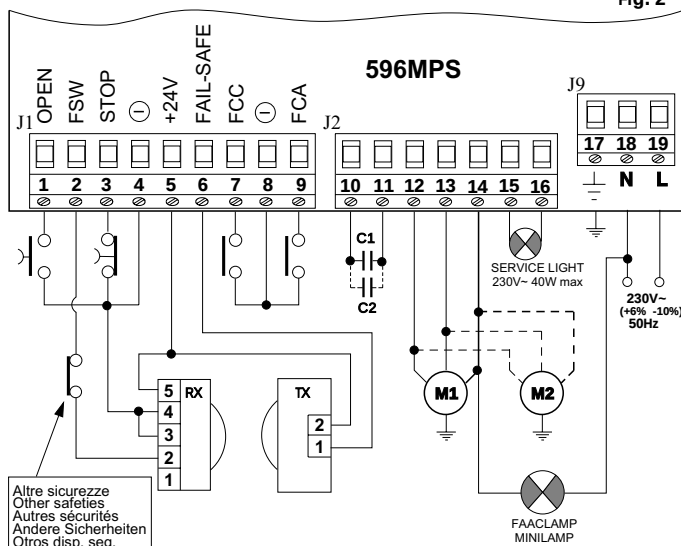
Before attempting any work on the card (connections, programming, maintenance), always turn off power.

Observe points 10, 11, 12, 13 and 14 of the GENERAL SAFETY RULES. Always separate 230V~ power cables from control and safety cables (push-button receiver, photocells, etc.).

To prevent any electric noise whatever, use separate sheaths.

2.1. 596MPS CONNECTION LAY-OUT (photocell with failsafe enabled)

Fig. 2



2.2. 610MPS CONNECTION LAY-OUT (photocell with failsafe enabled)

Fig. 3

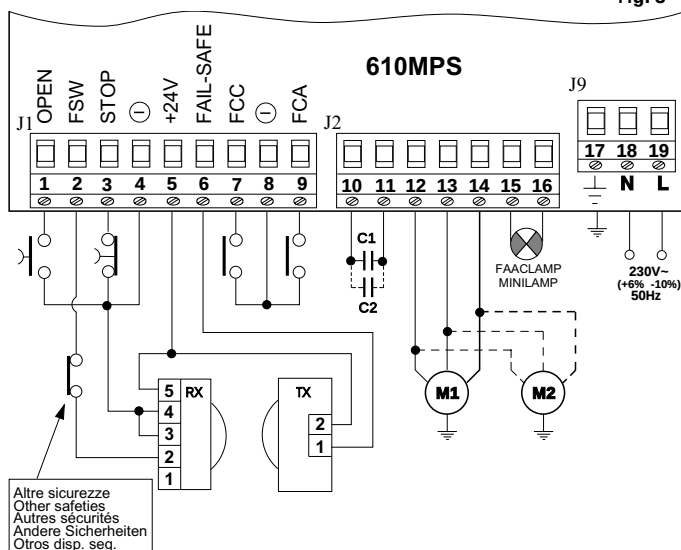


TABLE 3 COMPONENTS OF 596MPS & 610MPS CARD

F1	Fuse F1 5x20 5A/250V (motor/transformer primary winding)
F2	Fuse F2 5x20 0.5A/250V delayed (accessories power supply)
DS1	Programming microswitches
J1	Low voltage input/accessories terminal board
J2	Output terminal board for motor/capacitor/lamp
J3	Rapid connector for decoding/RP receiver cards
J4	Rapid connector for Open push-button
J9	230V power supply input terminal board

3. WIRING DESCRIPTION

3.1. TERMINAL BOARD J1 (low voltage)

1 = OPEN command (N.O.)

Any device (push-button, detector,...) which, by closing a contact, supplies an opening or closing pulse to the automatic system.

To install several Open devices, connect N.O. contacts in parallel.

2 = FSW Closing safety devices contact (N.C.)

Safety devices are all devices (photocells, sensitive edges, magnetic coils) with N.C. contact, which operate if there is an obstacle in the area they protect.

Safety devices protect the area affected by the closing movement of the automatic system.

If the safety devices are tripped during closure, automatic movement is reversed, whereas they have no effect during opening. The safety devices operating when the automatic system is open or pausing, prevent closure.

To install several safety devices, connect the N.C. contacts in series.

⚠ If no closing safety devices are installed, fit a jumper on this input:

- with terminal 4 (if failsafe is disabled), or
- with terminal 6 (if failsafe is enabled).

3 = STOP command (N.C.)

This is any device (e.g. a push-button) which, by opening a contact, stops movement of the automatic system.

To install several stop devices, connect the N.C. contacts in series.

⚠ If Stop devices are not connected, link the input to the common contact (terminal 4) via a jumper.

4 = Common for contacts/Negative for accessories supply (-)

5 = Positive for 24 Vdc accessories supply (+)

Max load of accessories: 250 mA.

To calculate absorption values, refer to the instructions for individual accessories.

6 = Failsafe (-) (see "4. Programming")

If Failsafe is enabled, you must connect the negative poles of the photocells projectors to this terminal.

7 = FCC Closing limit-switch contact (N.O.)

The closing limit-switch is a device with a N.O. contact which, by being tripped when the automatic system reaches the closed position (closing the contact), causes the automatic system to stop after about 1 second.

8 = Common for contacts/Negative for accessories supply (-)

9 = FCA Closing limit-switch contact (N.O.)

The opening limit-switch is a device with a N.O. contact which, by being tripped when the automatic system reaches the open position (closing the contact), causes the automatic system to stop.

3.2. TERMINAL BOARD J2 (high voltage)

10-11 = Connection to initial thrust capacitor

If the application has two operators, connect the capacitors in parallel.

12 - 13- 14 = Connection to electric motor

Terminals 12-13 should be connected to the motor phases (brown and black cables) whereas terminal 14 should be connected to the motor (blue cable).

If the application has two operators, connect the two motors in parallel.

15-16 = Timed courtesy light (596MPS)

Flashlight (610MPS)

The function of this output differs according to card model:

•596MPS:

Connect the courtesy light - 230V~ 60W max. - if any. The courtesy light is activated when the motor starts and stays lighted for about 90 seconds after end of movement.

•610MPS:

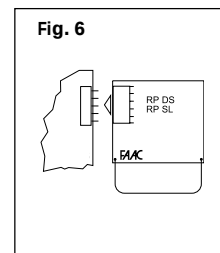
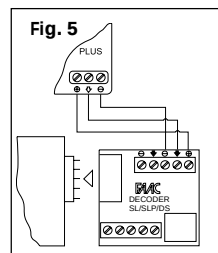
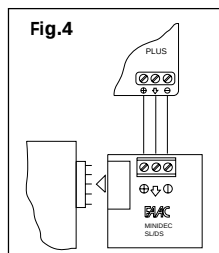
Connect the flashlight - 230V~ 60W max.

3.3. CONNECTOR J3 (low voltage)

Connector J3 is used for rapid connection of MINIDEC, DECODER, and RP RECEIVER cards (Fig. 4,5,6)

Install by fitting the accessory cards so that their component side faces the inside of the electronic appliance.

Insert and remove the cards after cutting power.



3.4. CONNECTOR J4 (low voltage)

The J4 connector is used for rapid connection of the opening push-button on the up-and-over operators' housing.

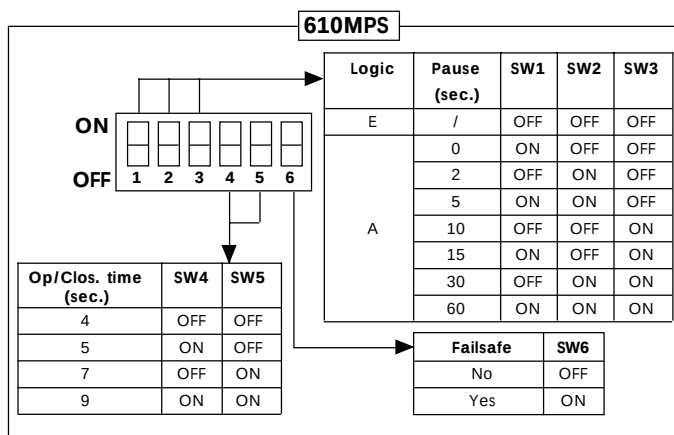
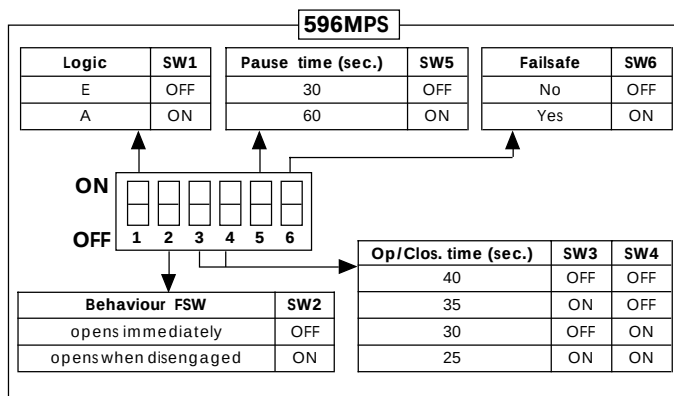
3.5. TERMINAL BOARD J9 (high voltage)

Terminal board for connecting 230V~ 50Hz power supply.

[L=Line - N=Neutral - W= Ground].

4. PROGRAMMING

To programme operation of the automatic system, turn the microswitches as follows:



•Function logics

Behaviour of the automatic system in logics:

A = Automatic E = Semi-automatic

is indicated on Tables 4-5.

•Pause time

Pause time in open state before re-closing, when the automatic logic is selected.

•Opening/Closing time

Select a movement time enabling the electric motor to receive power for a few seconds after the automatic system reaches the mechanical stops.

This is also the maximum time for reaching the limit-switches (optional items).

•Behaviour of FSW (596MPS only)

Defines the effect of the safety devices being tripped during closing (immediate reversing or reversing on disengagement).

Note: the 610MPS executes immediate reversing only.

•Failsafe

Failsafe is an electronic system for checking efficiency of the connected photo cells. The check is effected before each closing activation by the motors, by temporarily cutting power to the photocell projector (TX) to check the resulting opening of the contact on the receiver (RX). If this does not occur, the movement of the automatic system is disabled for reasons of safety. *To enable Failsafe, a separate power supply to the photocell projectors (Fig.12) is necessary.*

TABLE 4 AUTOMATIC LOGIC (A)

LOGIC "A"	PULSES		
AUTOMATION STATUS	OPEN	STOP	SAFETY DEVICES
CLOSED	opens and closes after pause time	no effect (Open inhibited)	no effect
OPENING	no effect	disabled operating	no effect
OPEN FOR PAUSE	re-closes immediately	disabled operating	stop pause until disengaged
CLOSING	re-opens immediately	disabled operating	reverses motion (★)
STOPPED	re-closes	no effect (Open inhibited)	no effect (Open inhibited)

TABLE 5 SEMI-AUTOMATIC LOGIC (E)

LOGIC "E"	PULSES		
AUTOMATION STATUS	OPEN	STOP	SAFETY DEVICES
CLOSED	opens	no effect (Open inhibited)	no effect
OPENING	stops	disabled operating	no effect
OPEN	re-closes immediately	no effect (Open inhibited)	no effect (Open inhibited)
CLOSING	re-opens immediately	disabled operating	reverses motion (★)
STOPPED	re-closes	no effect (Open inhibited)	no effect (Open inhibited)

(★) •596MPS=see SW2 •610MPS=reverses immediately

5. START-UP

Install the card by placing it in containers with an adequate degree of protection. The wiring accessories (cable sleeves, tubes...) must have the same degree of protection as the container.

5.1. ROTATION DIRECTION CHECK

- 1) Cut power to the card
- 2) Release the automatic system by using the operators lock/release device.
- 3) Manually move the automatic system halfway along its travel.
- 4) Re-lock the automatic system by using the operators lock/release device.
- 5) Restore power to the card.
- 6) Send an opening pulse (Open) and check if the automatic system is commanded to open.

If one or both motors are activated at closing, change over their phases (brown and black cables) on the card terminal board.

6. EXAMPLES OF SAFETY DEVICE CONNECTIONS

⇒ FAILSAFE disabled (SW6=OFF)

No safety device connected

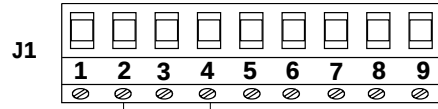


Fig. 7

Connection of a safety edge

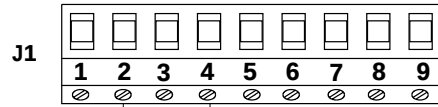


Fig. 8

Connection of a pair of photocells

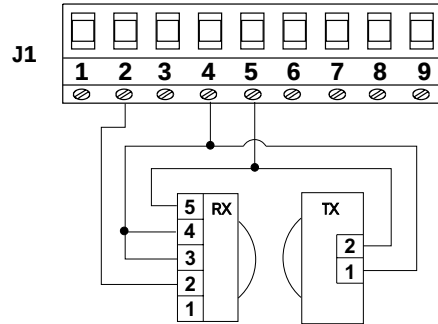


Fig. 9

7. EXAMPLES OF SAFETY DEVICE CONNECTIONS

⇒ FAILSAFE enabled (SW6=ON)

No safety device connected

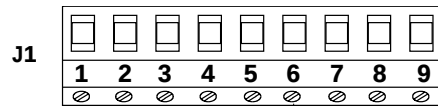


Fig. 10

Connection of a safety edge

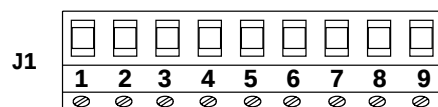


Fig. 11

Connection of a pair of photocells

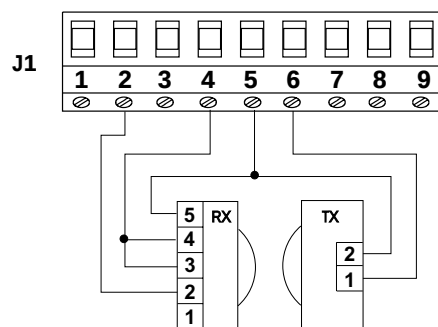


Fig. 12

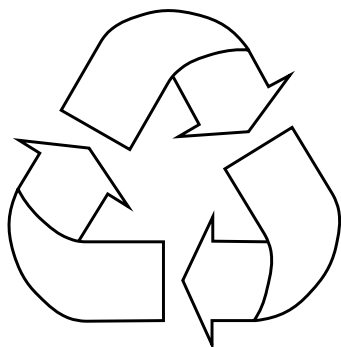
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- La presente istruzione è realizzata al 100% in carta riciclata.
- Non disperdete nell'ambiente gli imballaggi dei componenti dell'automazione bensì selezionate i vari materiali (es. cartone, polistirolo) secondo prescrizioni locali per lo smaltimento rifiuti e le norme vigenti.

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- The present manual is produced in 100% recycled paper
- Respect the environment. Dispose of each type of product packaging material (card, polystyrene) in accordance with the provisions for waste disposal as specified in the country of installation.

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